



PIXEL STUDIO

Image Processing

NAME

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REGISTRATION ID

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PROYECT PROPOSAL

In this project, we will use applied mathematics to create an image and video editing application, in addition to developing a color recognition function using the device's camera.

This application will be built in .NET Windows Forms in C#, using the Emgu CV library for the Video procesing and OpenCvSharp for the color capture feature. With these tools, we will develop an application that allows the following functionalities:

For both images and videos, the application will allow the user to select images or videos from their local device. The application will display the initial RGB image histogram. In the case of videos, this will change as the video progresses frame by frame.

After this, you can apply any filter of your choice and adjust some of them to your preference. You can also revert the change if desired. Similarly, when applying the filter, the histograms will update, showing the changes made after the filter is applied.

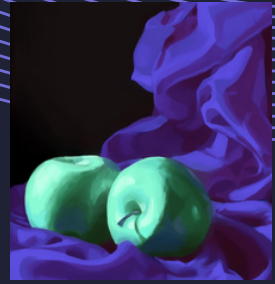
The camera will attempt to locate a color within the color space model and display its location in LA and B, as well as its hexadecimal code.

The proposed design for each of the required screens will then be displayed.

FILTERS

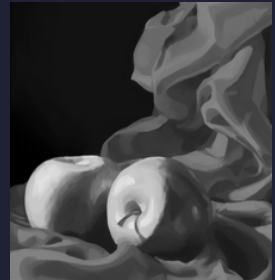
Inverted

This filter is achieved by inverting the RGB channels to change the original color to its complementary color on the color wheel. In this case, the Red channels are inverted for Blue and Blue for Red.



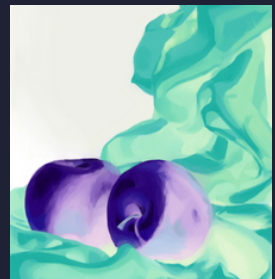
GrayScale

This filter will find the average of the values of each of the RGB fields and then assign that value to each of the RGB fields, thus creating a balance between the colors, converting the image into black and white.



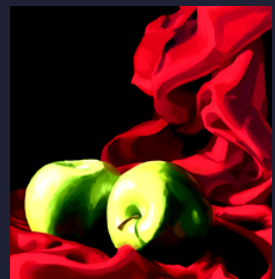
Negative

In this filter you will find the difference between the original color value and the limit of the color value (255) to obtain this difference, that is, if the color value of our pixel is 100 and the limit is 255, their difference will be 155, this value will be the one that will give our image this negative effect.



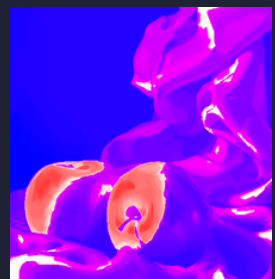
Contrast

This filter will adjust the colors to brighter and deeper values, first the intensity of this contrast will be determined, and in this way a formula will be used to adjust the pixel to the intensity of the contrast.



HeatMap

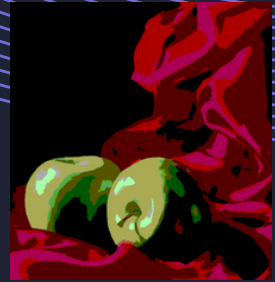
In this filter, the image will first be converted to grayscale to determine the intensity of each pixel. Following this, depending on this intensity, the colors between red and blue will be mapped, taking into account that this does not exceed the limits, that is, that it is between 0 and 255.



FILTERS

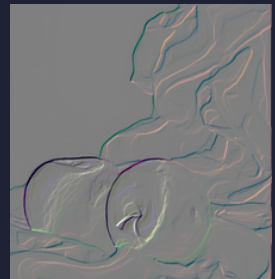
Posterize

This filter will reduce the amount of colors in an image by simplifying it into areas of the same color giving this cel shading effect, it takes a levels value and for each RGB field it will reduce the colors by dividing them based on the level value and assigning each pixel the closest color.



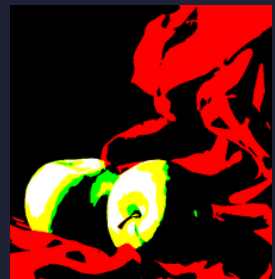
Emboss

This filter will be produced by comparing the current pixel with its neighbor, subtracting the position of the current pixel to take the next one in both x and y. The value of the subtraction will be what determines the thickness of this relief.



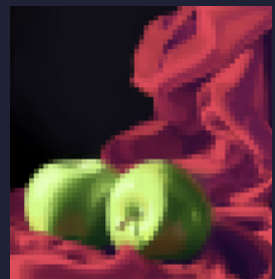
Comic

This filter will accentuate the lights and shadows, giving a lesser depth, resembling a style like comics or old prints. It will take the pixel color and take it to its maximum to bring all the colors to this effect.



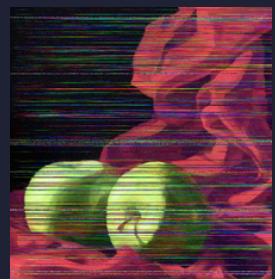
Pixel

This filter will take an initial value which will be the size of the pixel which will be used to separate the image into sets of n pixels each, followed by this all the colors found in each of the pixels within the set will be added and this will be the color that will be assigned to all the pixels in the previous set.



Glitch

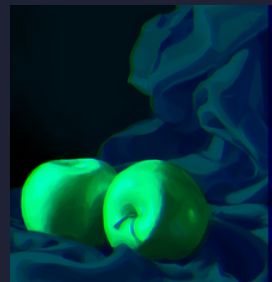
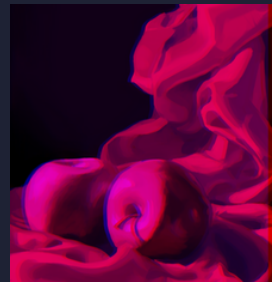
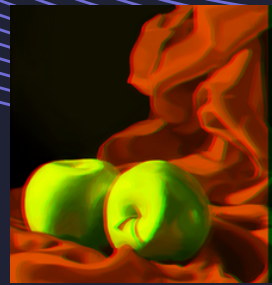
This filter will generate a digital glitch effect in the image by randomly shifting horizontal lines of color throughout it, altering the RGB channels in the same way and simulating noise in the image.



FILTERS

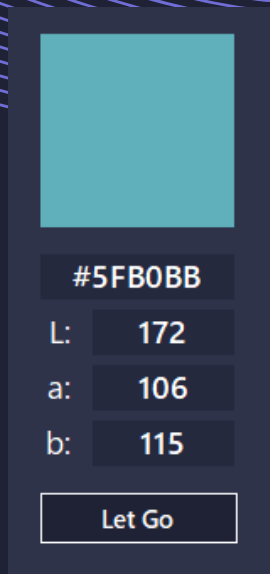
Glitch

The following glitch filters use the same logic; the only difference between them is which color channels are shifted and which are kept the same. This filter will be based on the intensity of the glitch, which will be the color shift. Depending on the filter, the color will be added and subtracted to create the color shift effect.



COLOR CAPTURE

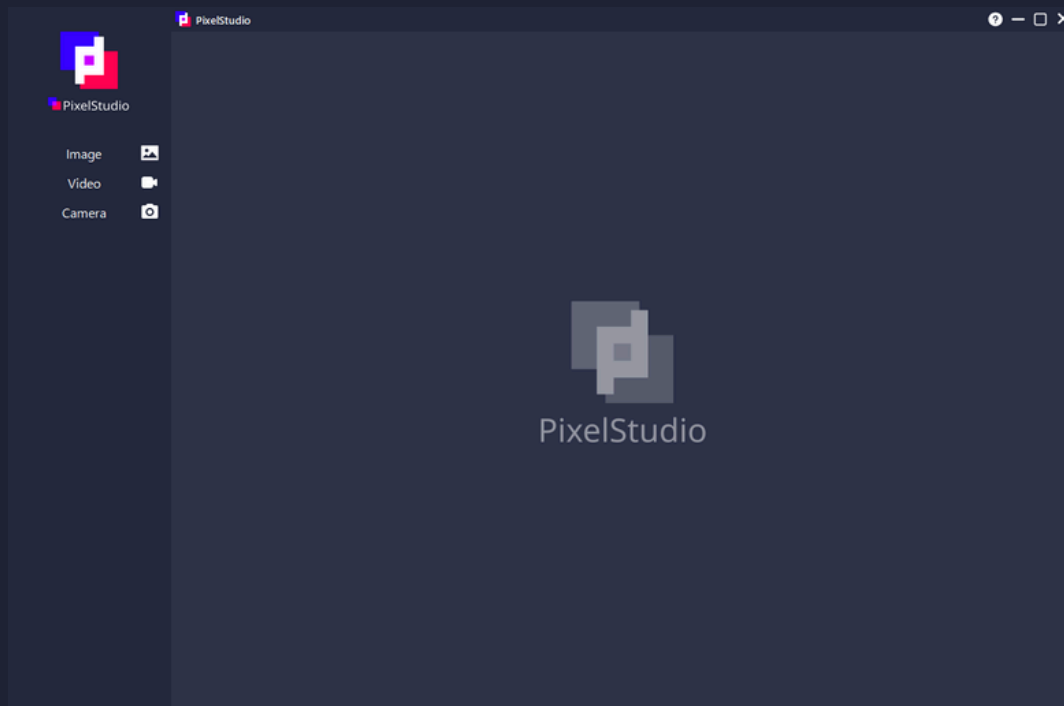
To be able to capture a color using the camera of our device the program will process each frame and get the color from the center of our image and convert those values to Hexadecimal and the values for CIELAB color space values (L, a, b). To then be able to display both the color and numeric values. The user will be able to see the color detection live and capture or freeze in the desired color.



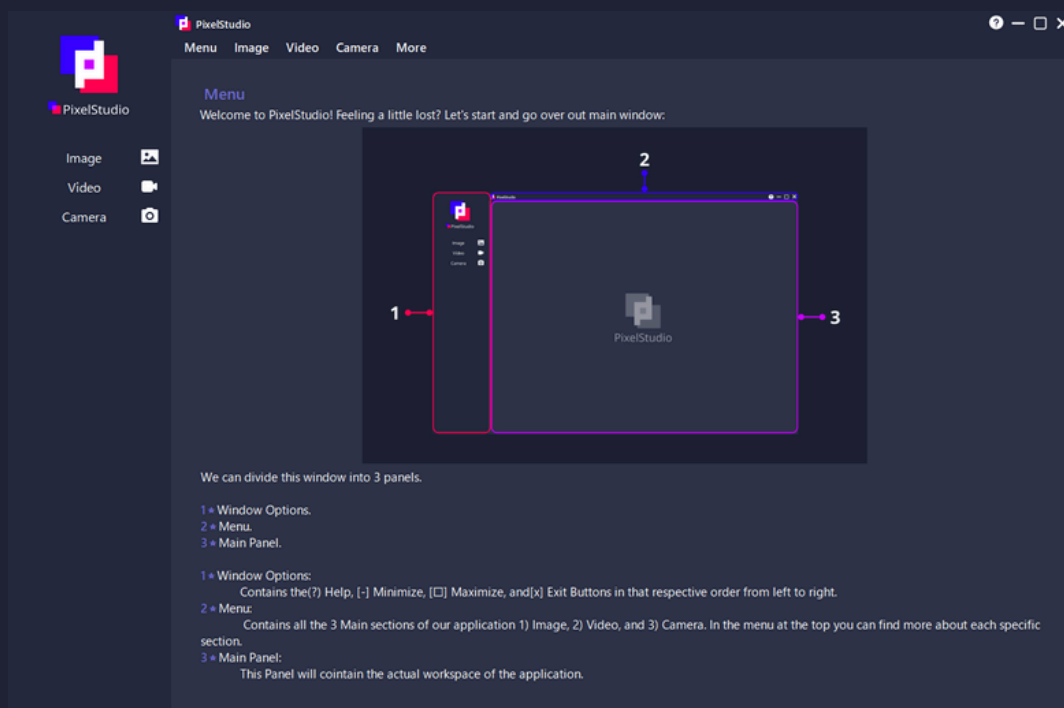
	
#5FB0BB	
L:	172
a:	106
b:	115
Let Go	

WINDOW DESING

inicio

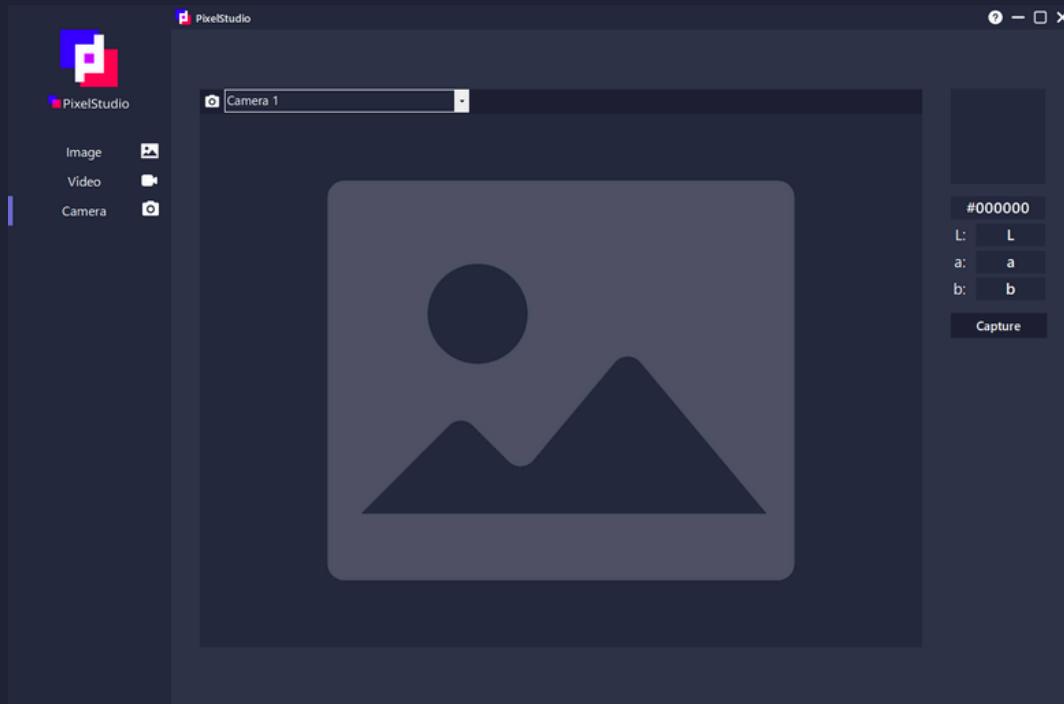


Manual de usuario

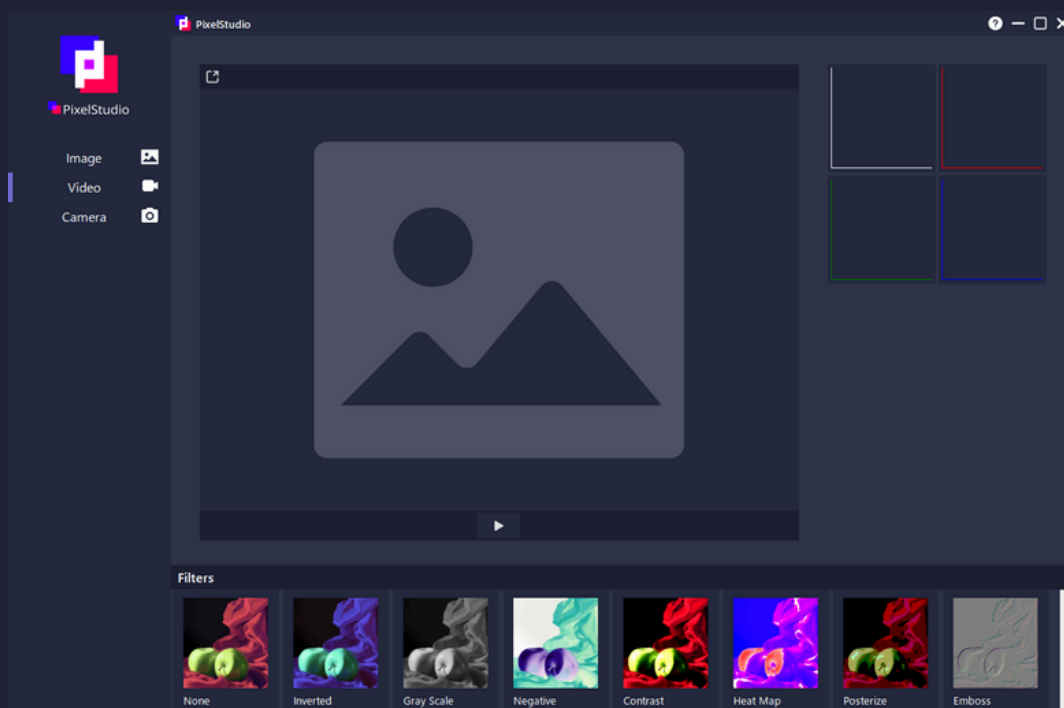


WINDOW DESING

Imagenes



Video



WINDOW DESING

Camara

